CSCI 49376 Assignment 1

Mia Elena Yábar (Partner A: design) · Jas (Partner B: implementation)
Big Data Technology · Prof. Alex Liew · Parts 1–4

Part 1. Select and Examine the Original Dataset

#	Required	Answer
1	Dataset selected	Option 2: Diabetes 130-US Hospitals for Years 1999–2008 (Kaggle: brandao/diabetes)
2	CSV file used	`diabetic_data.csv`
3	Total rows	101,766 (one row per hospital stay)
4	Total columns	50 (24 ignored as instructed, 26 used)

5–6. Attributes and what each represents

Attribute	Description
encounter_id	Unique ID of one hospital stay
patient_nbr	ID of a patient (a patient can have several stays)
race	Caucasian, AfricanAmerican, Hispanic, Asian, Other
gender	Female, Male, Unknown/Invalid
age	Age group in 10-year bands, e.g. [70-80)
weight	Weight band in pounds
admission_type_id	Code: type of admission (e.g. emergency, elective)
discharge_disposition_id	Code: where the patient went at discharge (e.g. home)
admission_source_id	Code: where the patient came from (e.g. referral, ER)
time_in_hospital	Days in the hospital (1–14)
payer_code	Insurance code (e.g. MC = Medicare)
medical_specialty	Specialty of the admitting doctor
num_lab_procedures	Lab tests during the stay
num_procedures	Other procedures during the stay
num_medications	Distinct medications given
number_outpatient	Outpatient visits in the prior year
number_emergency	Emergency visits in the prior year
number_inpatient	Inpatient visits in the prior year
diag_1	Primary diagnosis (ICD-9 code)
diag_2	Secondary diagnosis (ICD-9 code)
diag_3	Additional diagnosis (ICD-9 code)
number_diagnoses	Number of diagnoses recorded
max_glu_serum	Glucose test: >200, >300, Norm, None (not measured)
A1Cresult	HbA1c test: >7, >8, Norm, None (not measured)
diabetesMed	Any diabetes medication prescribed: Yes/No
readmitted	NO, >30 (after 30 days), <30 (within 30 days)

Attribute	Description
metformin, repaglinide, nateglinide, chlorpropamide, glimepiride, acetohexamide, glipizide, glyburide, tolbutamide, pioglitazone, rosiglitazone, acarbose, miglitol, troglitazone, tolazamide, examide, citoglipton, insulin, glyburide-metformin, glipizide-metformin, glimepiride-pioglitazone, metformin-rosiglitazone, metformin-pioglitazone, change	Ignored, as instructed: 23 medication columns (No, Steady, Up, Down) and whether any medication changed

7. Possible identifiers or keys

Column	Finding
encounter_id	Unique on all 101,766 rows → primary key of the original table
patient_nbr	71,518 distinct values; repeats across stays → identifies a separate PATIENT entity
admission_type_id, discharge_disposition_id, admission_source_id	8, 26 and 17 codes → reference tables (descriptions in `IDs_mapping.csv`, UCI)
diag_1, diag_2, diag_3	915 distinct ICD-9 codes → a DIAGNOSIS entity

8. Missing, unknown or incomplete values

Column	Missing	Written as
weight	96.9%	?
max_glu_serum	94.7%	None (not measured)
A1Cresult	83.3%	None (not measured)
medical_specialty	49.1%	?
payer_code	39.6%	?
race	2.2%	?
diag_3 / diag_2 / diag_1	1.4% / 0.4% / 21 rows	?
gender	3 rows	Unknown/Invalid

All are stored as NULL. Inconsistencies: 123 patients have two different race values and 3 have two genders across stays (we keep the value from the patient's latest stay); age changes between stays for 1,574 patients, so age belongs to the stay.

Part 2. Normalize the Dataset

Original relation → 1NF → 2NF. Each stage answers the six required items.

Original relation

1. **Structure:** DIABETIC_DATA(encounter_id, patient_nbr, race, gender, age, weight, admission_type_id, discharge_disposition_id, admission_source_id, time_in_hospital, payer_code, medical_specialty, num_lab_procedures, num_procedures, num_medications, number_outpatient, number_emergency,

number_inpatient, **diag_1, diag_2, diag_3**, number_diagnoses, max_glu_serum, A1Cresult, diabetesMed, readmitted). One table, one row per hospital stay.
2. **Key:** encounter_id
3. **Functional dependencies:** encounter_id → every other column · patient_nbr → race, gender · admission_type_id → description (and the same for discharge_disposition_id and admission_source_id)
4. **Problems:** diag_1, diag_2 and diag_3 are a **repeating group**: three columns holding the same kind of value. Patient race and gender repeat on every stay (30,248 extra copies).
5. **Change needed for 1NF:** replace the three diagnosis columns with one row per diagnosis, and add diag_position (1 = primary, 2 = secondary, 3 = additional).
6. **Normal form:** not in 1NF, because of the repeating group.

1NF relation

1. **Structure:** ENCOUNTER_1NF(encounter_id, diag_position, icd9_code, patient_nbr, race, gender, age, …, readmitted). One row per diagnosis.
2. **Key:** (encounter_id, diag_position). We use the position, not the code, because 5,492 stays list the same code twice.
3. **Functional dependencies:**
    - FD1: (encounter_id, diag_position) → icd9_code (needs the whole key)
    - FD2: encounter_id → patient_nbr, age, weight, the three codes, time_in_hospital, …, readmitted (needs only part of the key)
    - FD3: patient_nbr → race, gender
    - FD4: each code → its description
4. **Problems:** FD2 is a **partial dependency**. Every stay column is copied once per diagnosis (see the example below).
5. **Change needed for 2NF:** move FD2's columns into ENCOUNTER, keyed by encounter_id. To minimize redundancy, also move FD3 to PATIENT and FD4 to three code tables, and list each diagnosis code once in DIAGNOSIS.
6. **Why it is 1NF:** every cell holds one value, and the repeating group is gone.

Example: one stay with three diagnoses becomes three rows, and its time_in_hospital is copied each time (the partial dependency).

encounter_id	diag_position	icd9_code	time_in_hospital	…
1001	1	250.01	3	…
1001	2	401	3	…
1001	3	428	3	…

2NF relations

1. **Structure:** ENCOUNTER(encounter_id, patient_nbr, admission_type_id, discharge_disposition_id, admission_source_id, age, …, readmitted) · ENCOUNTER_DIAGNOSIS(encounter_id, diag_position, icd9_code) · PATIENT(patient_nbr, race, gender) · DIAGNOSIS(icd9_code) · ADMISSION_TYPE · DISCHARGE_DISPOSITION · ADMISSION_SOURCE (each: id, description)
2. **Keys:** listed in the final tables below.
3. **Functional dependencies:** FD1 in ENCOUNTER_DIAGNOSIS · FD2 in ENCOUNTER · FD3 in PATIENT · FD4 in the code tables
4. **Problems:** none. There are no partial dependencies, and each fact is stored once.
5. **Further changes:** none needed; this is the final design.
6. **Why it is 2NF:** every non-key column depends on the whole key. Tables with a one-column key can't have a partial dependency, and in ENCOUNTER_DIAGNOSIS, icd9_code needs both encounter_id and

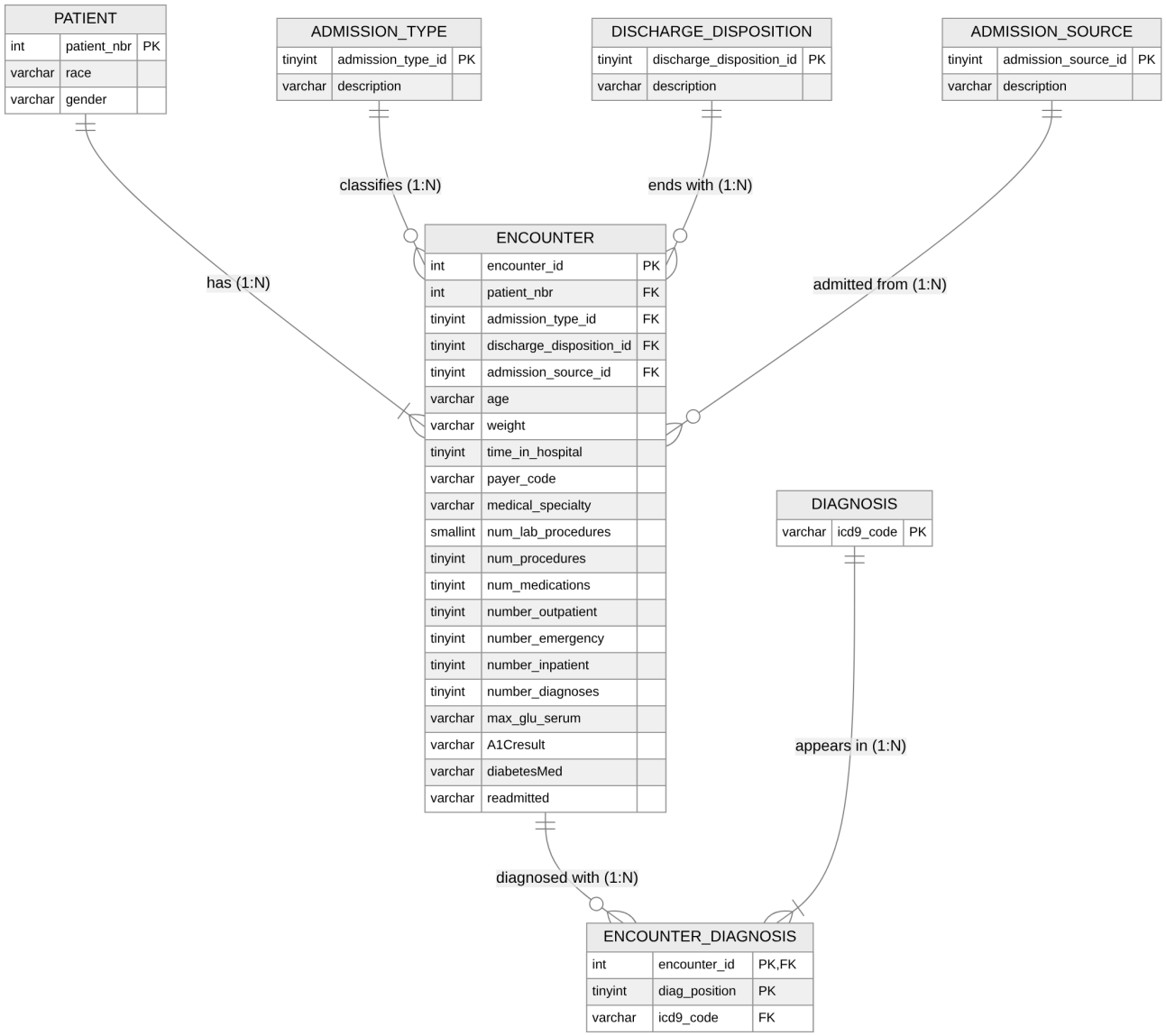
diag_position.

Final tables

Table	Attributes	Primary key	Foreign key(s)
PATIENT	patient_nbr, race, gender	patient_nbr	—
ENCOUNTER	encounter_id, patient_nbr, admission_type_id, discharge_disposition_id, admission_source_id, age, weight, time_in_hospital, payer_code, medical_specialty, num_lab_procedures, num_procedures, num_medications, number_outpatient, number_emergency, number_inpatient, number_diagnoses, max_glu_serum, A1Cresult, diabetesMed, readmitted	encounter_id	patient_nbr → PATIENT admission_type_id → ADMISSION_TYPE discharge_disposition_id → DISCHARGE_DISPOSITION admission_source_id → ADMISSION_SOURCE
ENCOUNTER_DIAGNOSIS	encounter_id, diag_position, icd9_code	(encounter_id, diag_position)	encounter_id → ENCOUNTER icd9_code → DIAGNOSIS
DIAGNOSIS	icd9_code	icd9_code	—
ADMISSION_TYPE	admission_type_id, description	admission_type_id	—
DISCHARGE_DISPOSITION	discharge_disposition_id, description	discharge_disposition_id	—
ADMISSION_SOURCE	admission_source_id, description	admission_source_id	—

Nothing is lost: joining the tables rebuilds every original row. Expected rows: ENCOUNTER 101,766 · PATIENT 71,518 · ENCOUNTER_DIAGNOSIS 303,496 · DIAGNOSIS 915 · code tables 8 / 26 / 17.

Part 3. Construct an ER Diagram



Notation (crow's foot): two bars = exactly one · bar + crow's foot = one or many (total participation) · circle + crow's foot = zero or many (partial participation). PK = primary key, FK = foreign key.

Relationships, cardinalities and participation

Relationship	Between	Type	Why	Participation
has	PATIENT – ENCOUNTER	1:N	A patient has many stays; a stay has one patient.	Total on both sides
classifies	ADMISSION_TYPE – ENCOUNTER	1:N	A type covers many stays; a stay has one type.	Total for ENCOUNTER, partial for the type
ends with	DISCHARGE_DISPOSITION – ENCOUNTER	1:N	Same pattern.	Total for ENCOUNTER, partial for the disposition

Relationship	Between	Type	Why	Participation
admitted from	ADMISSION_SOURCE – ENCOUNTER	1:N	Same pattern.	Total for ENCOUNTER, partial for the source
diagnosed with	ENCOUNTER – DIAGNOSIS	M:N	A stay has up to 3 diagnoses; a diagnosis appears in many stays. Drawn as two 1:N relationships through ENCOUNTER_DIAGNOSIS, which also stores diag_position.	Partial for ENCOUNTER (some stays have no diagnosis), total for DIAGNOSIS

There are no 1:1 relationships. The diagram matches Part 2: same seven tables, keys and foreign keys.

Part 4. Convert the ER Diagram into Relational Tables

Relation	Attributes	Primary key	Foreign key(s)
PATIENT	patient_nbr, race, gender	patient_nbr	—
ADMISSION_TYPE	admission_type_id, description	admission_type_id	—
DISCHARGE_DISPOSITION	discharge_disposition_id, description	discharge_disposition_id	—
ADMISSION_SOURCE	admission_source_id, description	admission_source_id	—
DIAGNOSIS	icd9_code	icd9_code	—
ENCOUNTER	encounter_id, patient_nbr, admission_type_id, discharge_disposition_id, admission_source_id, age, weight, time_in_hospital, payer_code, medical_specialty, num_lab_procedures, num_procedures, num_medications, number_outpatient, number_emergency, number_inpatient, number_diagnoses, max_glu_serum, A1Cresult, diabetesMed, readmitted	encounter_id	patient_nbr, admission_type_id, discharge_disposition_id, admission_source_id
ENCOUNTER_DIAGNOSIS	encounter_id, diag_position, icd9_code	(encounter_id, diag_position)	encounter_id, icd9_code

How the ER relationships are represented

ER relationship	Type	Represented as
PATIENT has ENCOUNTER	1:N	Foreign key ENCOUNTER.patient_nbr → PATIENT (on the N side, NOT NULL)
ADMISSION_TYPE classifies ENCOUNTER	1:N	Foreign key ENCOUNTER.admission_type_id → ADMISSION_TYPE
DISCHARGE_DISPOSITION ends with ENCOUNTER	1:N	Foreign key ENCOUNTER.discharge_disposition_id → DISCHARGE_DISPOSITION
ADMISSION_SOURCE admitted from ENCOUNTER	1:N	Foreign key ENCOUNTER.admission_source_id → ADMISSION_SOURCE
ENCOUNTER diagnosed with DIAGNOSIS	M:N	Its own table, ENCOUNTER_DIAGNOSIS: a foreign key to each side plus diag_position

Table creation order: PATIENT, ADMISSION_TYPE, DISCHARGE_DISPOSITION, ADMISSION_SOURCE, DIAGNOSIS → ENCOUNTER → ENCOUNTER_DIAGNOSIS.